

1. Linearly separable:

- A. in 2D space, the data points can be separated by a straight line.
- B. in 3D space, the data points can be separated by a flat plane.

2. limitation of a single perceptron:

- A. a single perceptron can **only** solve **linearly separable** issues.
- B. it's impossible to separate **XOR Gate** data points with a straight line in 2D space, but several straight lines is available.
- C. in other words, it's impossible to solve **XOR Gate** issues with a single perceptron, but combining multiple perceptrons is available.

D.

Table 1-2 Truth Table for a Two-Input XOR Gate

x_0	x_1	y
False	False	False
True	False	True
False	True	True
True	True	False

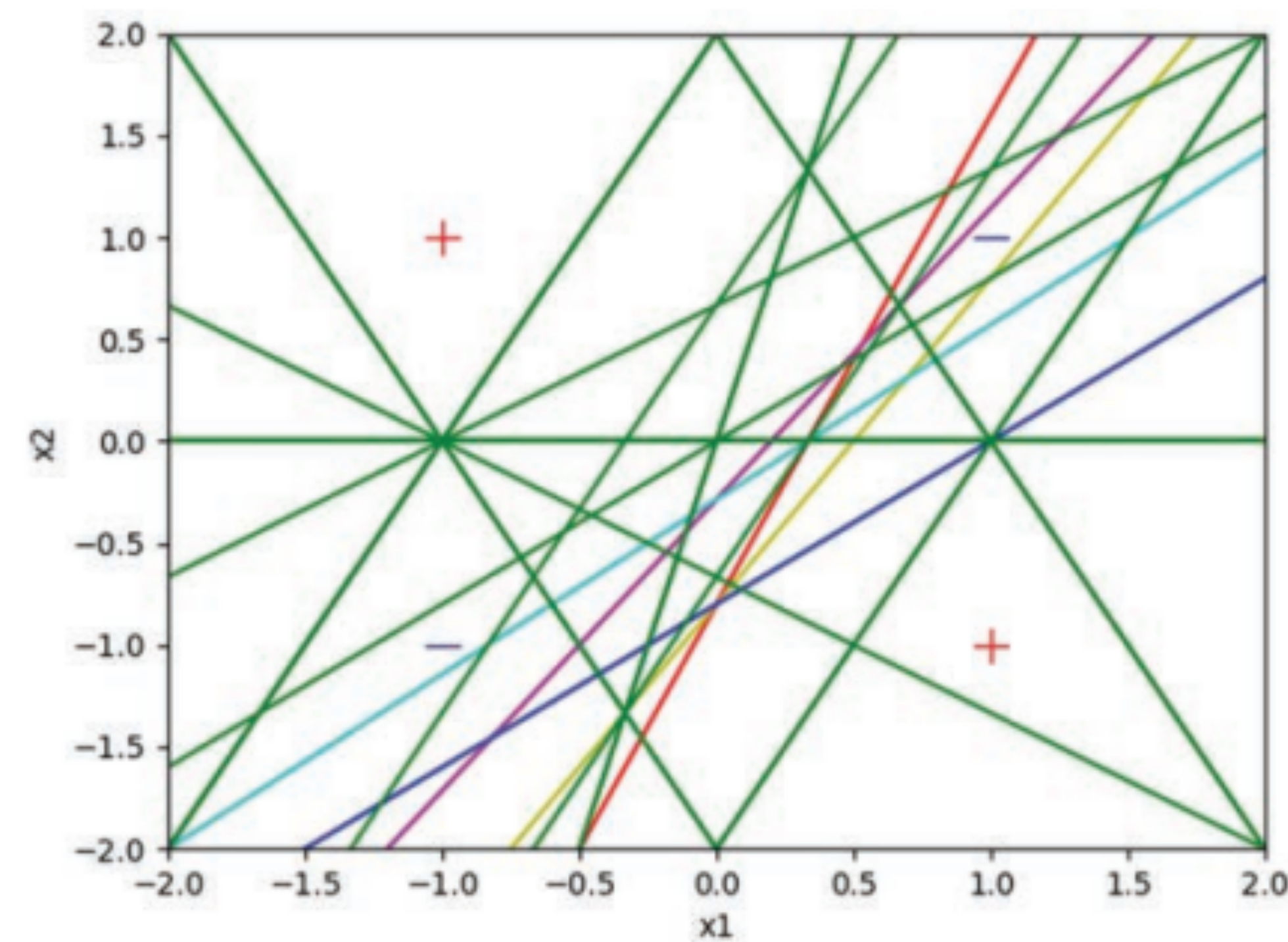
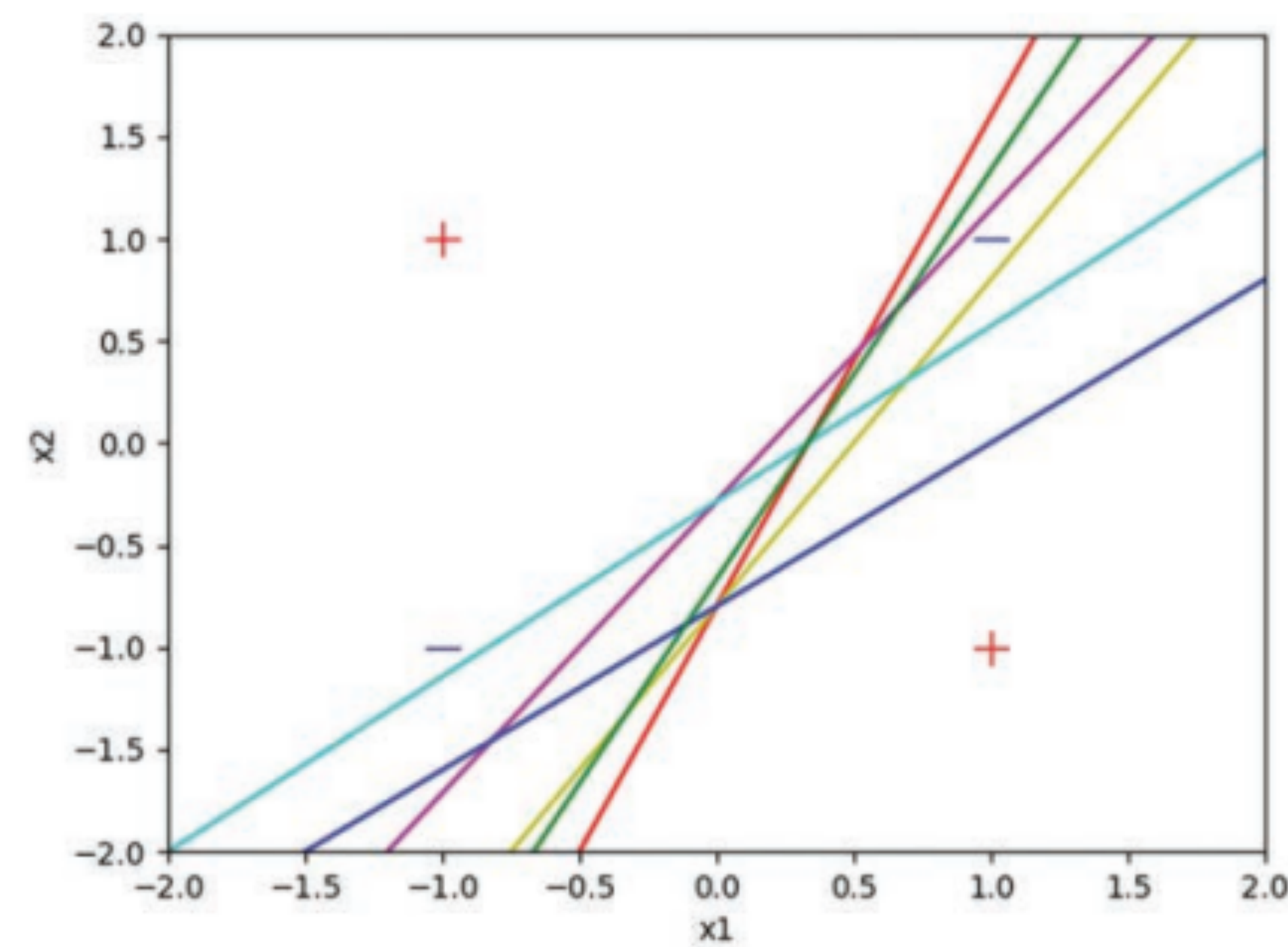


Figure 1-7 Perceptron attempting to learn XOR. Top: After 6 weight adjustments. Bottom: After 30 weight adjustments.

the result above shows that a single perceptron using learning algo never make the learning algo converge in a **XOR Gate** issue

3. learning algorithm converge: learning algorithm successfully finds out **fixed set of weights for perceptrons** to correctly match pair training example(s).